

Basics of Geometry

US Common Core – High School Geometry

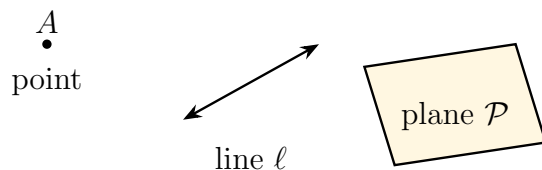
Domain: Congruence (G-CO.A.1) | Expressing Geometric Properties with Equations (G-GPE.B)

1 Points, Lines, and Planes

Undefined Terms

Geometry begins with three undefined terms:

- A **point** has no size – only a location. It is named with a capital letter.
- A **line** extends without end in two directions. It is named by any two points on it, or a single lowercase letter.
- A **plane** is a flat surface that extends without end in all directions. It is named by three points on it, or a single capital script letter.



Collinear and Coplanar

Points that lie on the same line are **collinear**. Points (or lines) that lie in the same plane are **coplanar**.

2 Segments, Rays, and Postulates

Definitions

A **line segment** $\overline{AB}$ consists of two endpoints A, B and all points between them. A **ray** $\overrightarrow{AB}$ starts at A and extends through B without end.

Ruler Postulate

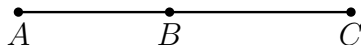
The points on a line can be matched, one to one, with the real numbers. The distance between two points equals the absolute value of the difference of their coordinates:

$$AB = |x_2 - x_1|.$$

Segment Addition Postulate

If B lies between A and C on a line, then

$$AB + BC = AC.$$



3 Midpoint and Distance Formulas

Midpoint Formula

The midpoint M of the segment joining (x_1, y_1) and (x_2, y_2) is

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Distance Formula

The distance between (x_1, y_1) and (x_2, y_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Example

Find the midpoint and distance between $A(1, 2)$ and $B(7, 10)$:

$$M = \left(\frac{1 + 7}{2}, \frac{2 + 10}{2} \right) = (4, 6)$$

$$d = \sqrt{(7 - 1)^2 + (10 - 2)^2} = \sqrt{36 + 64} = \sqrt{100} = 10.$$

4 Perimeter and Area in the Coordinate Plane

Method

To find the perimeter of a polygon plotted on the coordinate plane, use the distance formula on each side and add the results. To find its area, count grid squares, decompose the figure into simpler shapes (triangles, rectangles), or apply a known area formula once the side lengths and/or coordinates are known.

5 Measuring and Constructing Angles

Definition

An **angle** consists of two rays (the sides) that share a common endpoint (the **vertex**). An angle is measured in degrees.

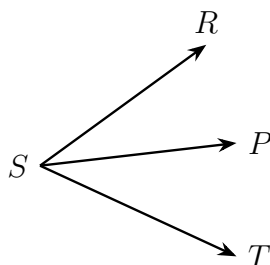
Protractor Postulate

Given a ray $\overrightarrow{AB}$ and a point A on line ℓ , all the rays drawn from A can be matched one to one with the real numbers from 0 to 180.

Angle Addition Postulate

If P is in the interior of $\angle RST$, then

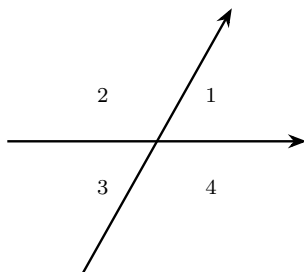
$$m\angle RST = m\angle RSP + m\angle PST.$$



6 Pairs of Angles

Definitions

- **Complementary angles:** two angles whose measures sum to 90° .
- **Supplementary angles:** two angles whose measures sum to 180° .
- **Linear pair:** two adjacent angles that form a straight line (supplementary and adjacent).
- **Vertical angles:** two angles formed by intersecting lines, opposite each other – always congruent.



Note: Angles 1 and 3 are vertical angles, and so are angles 2 and 4 – each pair is congruent. Angles 1 and 2 (or 2 and 3, etc.) form a linear pair and are supplementary.

Example

Two complementary angles measure x° and $(2x + 15)^\circ$:

$$x + 2x + 15 = 90 \Rightarrow 3x = 75 \Rightarrow x = 25^\circ.$$